

Machine Learning System for Intrapartum Fetal Distress Detection

Candidate: Devanshu Dhoble | Role: AI/ML Engineer | Organization: Janitri

1. How I Framed It

Problem Formulation & Clinical Context

During labour, uterine contractions (UC) periodically compress the placental vessels, temporarily diminishing maternal-fetal blood and oxygen exchange. A healthy fetus possesses physiological reserve and recovers between contractions. However, when oxygen delivery is persistently inadequate (due to cord compression, placental insufficiency, or hyperstimulation), the fetus transitions to anaerobic metabolism, leading to lactic acid accumulation and progressive metabolic acidosis. Clinicians monitor Cardiotocography (CTG) - tracking Fetal Heart Rate (FHR) and Uterine Contractions (UC) simultaneously - to identify early physiological signs of hypoxia.

I framed this task as a supervised binary classification problem: given the continuous FHR and UC time-series signals during active labour, predict whether the infant will suffer from intrapartum distress/hypoxia at delivery.

Defining 'Fetal Distress' from Delivery Outcomes

Delivery outcomes in the CTU-CHB database provide both biochemical measurements (umbilical cord blood gas: pH, BDecf, BE) and clinical vitality assessments (Apgar scores at 1 and 5 minutes). I defined a composite label:

Distressed ($y = 1$) if: (Umbilical Cord pH < 7.20) OR (5-minute Apgar Score < 7)

Not Distressed / Normal ($y = 0$): All other cases with valid measurements.

Rationale for this threshold:

- **Biochemical Basis (pH < 7.20)** : International obstetrics guidelines (ACOG, FIGO) establish pH < 7.20 as the clinical threshold for mild-to-significant intrapartum acidemia. Severe acidosis is marked at pH < 7.05-7.10, but 7.20 serves as the primary alert threshold where cellular oxygen debt begins.
- **Clinical Depressed State (Apgar5 < 7)** : The 5-minute Apgar score measures post-delivery physiological recovery (heart rate, respiration, muscle tone, reflexes, skin color). A score below 7 signifies neonate depression requiring resuscitation.
- **The Composite 'OR' Rule** : Combining biochemical and clinical markers ensures we capture both silent acidotic infants (low pH but compensated Apgar) and clinically depressed neonates. On the 552-record dataset, this produces 182 distressed cases (33.0%) and 370 normal cases (67.0%), providing sufficient positive examples for balanced machine learning.

Input Time Window Strategy

Labor recordings range from 40 minutes to multiple hours, but all recordings terminate at delivery. Because hypoxia and metabolic acid accumulate progressively, the terminal stage of labor contains the most discriminative physiological signals. I standardized the input window to the final 30 minutes before delivery (7,200 time steps at 4 Hz) for all recordings.

2. What I Built and How I Checked It

Signal Preprocessing & Feature Engineering

Ultrasound fetal heart rate monitoring suffers from acoustic loss and maternal movement artefacts. I designed a robust preprocessing pipeline: FHR values ≤ 50 bpm, ≥ 250 bpm, or equal to 0 were masked as NaN dropouts. Rather than relying on black-box deep learning on a modest 552-patient dataset, I engineered 29 domain-specific clinical features across six physiological categories:

- **Baseline & Distribution** : FHR mean, standard deviation, median, range, interquartile range (IQR), skewness, kurtosis, and rolling baseline heart rate.
- **Heart Rate Variability (HRV)** : Root Mean Square of Successive Differences (RMSSD), SDNN, Short-Term Variability (STV proxy over 1-min epochs), and Long-Term Variability (LTV proxy).
- **Decelerations & Accelerations** : Clinically defined deceleration count (FHR drop ≥ 15 bpm for ≥ 15 s) and acceleration count relative to estimated baseline.
- **Signal Reliability** : FHR missing ratio (artefact/dropout fraction) to measure monitoring stability.
- **Uterine Activity & Regularity** : UC mean, std, max amplitude, contraction frequency per minute, mean contraction interval, and interval standard deviation via peak prominence detection.
- **FHR-UC Coupling Dynamics** : Pearson cross-correlation between FHR and UC signals, and mean FHR during contractions vs. between contractions (quantifying late deceleration trends).

Model Architecture & Training Setup

I selected a Random Forest Classifier (300 estimators, max depth 12) combined with a StandardScaler. Key justifications: (1) Handles non-linear feature interactions without overfitting small sample sizes; (2) Inherently robust to multi-scale features; (3) Employs 'class_weight=balanced' to penalize misclassification of the distressed minority class; (4) Provides intrinsic feature importance rankings for medical explainability. The dataset (552 cases) was split 80/20 into 441 training samples and 111 held-out test samples using stratified sampling.

Test Set Evaluation & Metric Analysis

The model was evaluated on the independent 111-sample test set. The exact test performance metrics obtained are:

ROC-AUC: 0.7330	Accuracy: 70.27%	Specificity: 87.84%	Precision: 59.09%
Recall (Sens.): 35.14%	F1-Score: 0.4407	Confusion Matrix: TP=13, FP=9, FN=24, TN=65	
What the metrics tell us vs. what they do NOT tell us:			

- **ROC-AUC (0.7330)** : Tells us the model possesses solid, generalizable ranking capability across diverse probability thresholds (significantly above 0.50 random guessing). However, it does not reveal the operational performance at a specific clinical alert cut-off.
- **Specificity (0.8784)** : Tells us that 88% of healthy fetuses are correctly classified without false alerts. This is critical to prevent clinician alarm fatigue and avoid unnecessary emergency Caesarean deliveries.
- **Precision (0.5909)** : Indicates that nearly 60% of triggered alarms represent true distress cases.
- **Recall Gap (0.3514)** : Shows that at the default 0.50 decision threshold, the model catches 13 of 37 distressed fetuses, missing 24. In clinical practice, missing a compromised fetus is catastrophic; this demonstrates that a fixed 0.50 threshold is clinically unsuitable and must be calibrated downward to prioritize sensitivity.

3. Clinical Utility and Inference

Real-World Bedside Usage Scenario

This tool is designed as an Intelligent Bedside Decision Support System (CDSS) for labour and delivery wards. It does not replace obstetricians or midwives; rather, it provides an automated, continuous, objective 'second pair of eyes'.

- **Who monitors the output?** : Staff nurses, resident medical officers, and attending obstetricians in the labor ward.
- **When does it run?** : Continuously during active labor. Every 5 to 10 minutes, a background sliding window ingests the most recent 30 minutes of streaming CTG data and updates a real-time Risk Trajectory on the bedside monitor.
- **What does it return?** : (1) Continuous Distress Risk Index (0.0 to 1.0); (2) Visual Risk State (Green = Normal, Amber = Observe, Red = Urgent Review); (3) Top contributing feature factors (e.g. 'Repetitive Decelerations' or 'Depressed STV').

Alignment of generate_outcomes.py with Clinical Reality

The standalone inference script (generate_outcomes.py) precisely reflects this operational architecture. It decouples training from inference, accepting either raw WFDB streaming channels or pre-extracted feature vectors, applying the frozen StandardScaler transformation and Random Forest inference, and returning probability scores alongside configurable threshold-based alerts. This ensures zero data leakage and seamless integration into edge-device CTG hardware.

4. Limits and Next Steps

Honest Appraisal of Weaknesses and Failure Modes

- **Modest Sample Size (552 cases)** : 552 recordings collected from a single university hospital (CTU-CHB) limits physiological diversity and risks population overfitting. Validation across multi-center cohorts is essential.
- **Static Summary vs. Temporal Evolution** : Aggregating 30 minutes into a static 29-feature vector loses sequential temporal dynamics (e.g., whether decelerations are worsening or recovering over consecutive contractions).
- **Retrospective Delivery Ground Truth** : Umbilical cord pH and Apgar scores are measured post-delivery. An infant showing transient distress at minute -30 might recover before birth, leading to apparent 'false positive' labels.
- **Signal Artefacts in Active 2nd Stage** : During maternal pushing, ultrasound probe displacement causes extended dropouts, degrading HRV feature fidelity.

Concrete Engineering Roadmap (What I Would Try with More Time)

- **1. Dynamic Risk Trajectory & Sliding Windows** : Replace static single-window evaluation with continuous temporal risk tracking over consecutive 10-minute intervals.
- **2. Temporal Deep Learning Architectures** : Train 1D Convolutional Neural Networks (1D-CNN) paired with Bi-directional LSTMs directly on multi-channel raw FHR+UC signals to learn automated spatial-temporal representations.
- **3. Clinically Calibrated Thresholds & Platt Scaling** : Tune operational alert thresholds to achieve $\geq 80\%$ recall (sensitivity), utilizing cost-sensitive learning matrices to penalize false negatives heavily.
- **4. Explainable AI via SHAP** : Integrate TreeSHAP to display real-time feature attribution directly on the nurse's monitor (e.g., 'Warning: Risk elevated by 32% due to late decelerations').